

The Application Note Template

GFDL-1.3-or-later

This note is the template for the others. It is a working document: everything the collection uses is demonstrated here once, in place, so a writer starting a new note can copy this file, delete the prose, and keep the furniture. It is also the reference for the design language. When two notes disagree about how to draw a keystroke or where to put a caption, this one settles it.

The material divides in two. Sections 1 to 2.2 are about the mechanics: what to copy, what to rename, how to build, and what the page is made of. Sections 3 to 14 walk the constructs themselves, each shown as the source you write and the result you get.

Read Section 1 first and then skim for the construct you need. Nothing here has to be read end to end.

Numerical results in this document are of three kinds, and this paragraph is itself part of the template. Results quoted from a reference are marked as such. Results computed by hand are given with their working, so the reader can check them. Results the C47 test suite asserts are identified by the case that asserts them. Anything that has to be read off real hardware carries a marker in red. Keep this paragraph, edited to fit, in any note that quotes numbers. It tells the reader what a figure is worth.

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REV	DATE	DESCRIPTION
0.1	2026-07-29	First draft. Consolidates the keypress and screen notation of AN0029, AN0030 and AN0031 into <code>style/c47appnote.sty</code> , moves the body to Helvetica, and adds callout blocks, listing styles and the shared keyboard maps.

1 Using This Template

1.1 Copy, rename, gut

Three steps.

1. Copy this directory to `docs/appnotes/sources/AN00NN/` and rename the `.ltx` to match the convention in Section 1.2.
2. Edit the four front-matter lines in the preamble. They are the only place the note's number, title and date appear, and everything else follows from them.
3. Delete every section below and write the note. Keep the revision table, the bibliography, the keyboard appendix if the note needs it, and the author block.

The preamble you are keeping is four lines long:

```
\documentclass[canadian]{article}
\usepackage{c47appnote}

\notenum{032}
\noteshorttitle{Template}
\notedate{2026-07-29}
\notetitle{The Application Note Template}
```

All four feed the masthead and the running heads, so keep the short title short: it has to sit in the middle of a 7.4 pt header between the number and the date.

There is no byline in the masthead and no field for one. The author is named once, in `authorblock` at the back beside the colophon, which is where a reader looks for attribution and licensing together. A name

under the title would only repeat it.

NOTE

Everything the design language provides comes from one file, `sources/style/c47appnote.sty`, and that file is the single copy. Before AN0032 the same 50 lines of palette and keybox definitions were pasted into AN0029, AN0030 and AN0031, with a comment in each saying to consolidate them once all three had landed. This is that consolidation. Fix a macro once and every note gets the fix.

1.2 The filename convention

Two names, and they do not match each other. The source keeps the long hyphenated form, and the published PDF keeps the older spaced form the collection has always used.

source	docs/apnotes/sources/AN00NN/ AN-00NN-C47-R47-Application-Note-YYYY-MM-DD-XX-short-slug.ltx
PDF	docs/apnotes/ AN00NN YYYY-MM-DD C47 R47 XX Short Title.pdf

XX is the author's initials, and getting them wrong credits the work to somebody else. The map:

Table 1 Author initials as they appear in the collection

INITIALS	AUTHOR	NOTES
JM	Jaco Mostert	The bulk of the collection
DL	Didier Lachieze	AN0021 Printing, AN0026 42S Compatibility
DA	David Abram	AN0027 Structural conversions
WGR	Walter	AN0023 Elliptic integrals
NUK	Nigel UK	AN0000 Installation
DE	David Emerson	AN0029 First Steps, AN0030 Uncertainty, AN0031 Algebraic Numbers

WARNING

There are two Davids. DA is David Abram and DE is David Emerson. AN0030 and AN0031 were both filed as DA and had to be corrected. Never copy the initials from a neighbouring note's filename; check who wrote that one first.

The date in both names is the date of the revision being published, not the date the note was started. Bump it when you publish, and add a row to the revision table at the same time.

1.3 Building

The style file lives one level up from the note, so TEXINPUTS has to point at it. From the note's own directory:

```
TEXINPUTS=../style: lualatex AN-00NN-...-slug.ltx
TEXINPUTS=../style: lualatex AN-00NN-...-slug.ltx
```

Twice, so the table of contents, the lastpage count and the cross references settle. The trailing colon on TEXINPUTS matters: it is what keeps the default search path.

lualatex is required, not optional. The package calls `RequireLuaTeX` and will stop under `pdflatex` or `xelatex`.

2 The Page

2.1 Typography

Body and headings are TeX Gyre Heros, which is the metric-compatible Helvetica that ships with TeX Live. Using the clone rather than the system Helvetica is deliberate: every contributor gets the same page, including the ones on Linux and the CI runner, and nothing depends on what happens to be installed.

The page is letter with a 1.25 in margin, which puts the measure near 75 characters. That is the usual readability target, and it also leaves the keyboard maps room to sit at close to their natural width instead of being shrunk to fit.

Mathematics is New Computer Modern Sans Math, also from TeX Live, so a variable in a derivation matches the prose around it instead of falling back to a serif face. Compare the sum in Equation 1 with the sentence carrying it. Fira Math was tried first and rejected: its minus sign is undersized, so $n - 1$ reads as an en dash, and its ν is indistinguishable from a v , which matters in any note that uses ν for degrees of freedom.

One typographic accent does all the labelling: `\lsc`, a letterspaced uppercase run. It marks the masthead, a table head, a caption label and a callout title, so structural furniture never needs a colour or a rule of its own to be read as furniture.

The display face is JuliaMono, and it is the one font the package cannot supply. The screen and program boxes reproduce the calculator display and need its glyphs literally: $x=y$, x^2 , $\Sigma+$, $\sqrt{}$, $\neq$. Install it from <https://juliamono.netlify.app>. Without it the package falls back to DejaVu Sans Mono and warns during the build, and the boxes will be missing glyphs.

Table 2 The three faces and what each is for

ROLE	FACE	USED FOR
Body	TeX Gyre Heros	Prose, headings, tables, captions, keybox labels
Math	NewCM Sans Math	Everything between dollar signs
Display	JuliaMono	screen, prog, cmd, listings: anything quoting the calculator or the source tree

2.2 The palette

Seven colours, and two of them are spoken for.

Table 3 The palette. `fcol` and `gcol` carry a fixed meaning and are never available for anything else.

NAME	HEX	MEANING
keyink	 1B1B1B	Body, headings, rules, keyboard: nearly everything
soft	 6B6A65	Furniture and captions, and anything deliberately quiet
fcol	 DE6E10	The f shift, and everything reached through it
gcol	 2C6DB5	The g shift, and everything reached through it
menufill	 E9E7E2	The fill of a soft-key label
lcdbg	 F2F3EA	The paper colour of the display
warnink	 B3341F	The one accent: a warning, and the draft markers

Three rules hold the design together, and they are worth stating plainly because breaking any of them is what makes two notes stop looking like each other.

One. `fcol` orange means “reached through the f shift” and `gcol` blue means “reached through the g shift”, everywhere, without exception. Nothing else may use them: not a heading, not a rule, not a running head, not a callout. A reader who has learned that on page 2 must not meet an orange mark on page 9 that means something else. This is why the running heads are `soft` grey rather than the blue AN0029 through AN0031 first used.

Two. A filled panel is the calculator’s own screen. `screen` and `prog` carry the `lcdbg` wash because they reproduce the display; tables, listings and callouts are unfilled and bounded by hairlines instead. So a fill on the page always means the same thing, and a quotation of the calculator never looks like a quotation of a file.

Three. Rules carry weight in one direction only. 1 pt closes the masthead and opens the author block, 0.7 pt opens and closes a table, and 0.4 pt is every other hairline: the section rules, the header rule, a capture frame, a listing. No double rules and no vertical rules anywhere.

Note what is missing. There is no hue for a note or a tip, because their titles tell them apart and that is cheaper than teaching the reader two more colours. `warnink` is the only accent, spent on the one callout that is a safety signal and on the draft markers of Section 12. Everything else is `keyink`, `soft`, or a tint of `keyink`.

3 Keypress Notation

This is the part of the design language worth most, because it is the part a reader learns once and then reads without effort.

3.1 The kinds of key

Six macros, one per way of reaching a function.

Table 4 Keypress macros

SOURCE	RESULT	MEANS
<code>\key{7}</code>		Press that key
<code>\fkey{MODE}</code>		f shift, then the key whose orange label is MODE
<code>\gkey{CLK}</code>		g shift, then the key whose blue label is CLK
<code>\mkey{DEG}</code>		A soft key, named by the label the screen prints above it
<code>\fmkey{U.P}</code>		f shift, then the soft key carrying U.P on its orange row
<code>\gmkey{REGR}</code>		g shift, then the soft key carrying REGR on its blue row
<code>\lkey{f}</code>		A long press

On the C47 there is one physical shift key, pressed once for f and twice for g. On the R47 there are two, a yellow f and a blue g. The notation is the same on both, and a note giving C47 keystrokes need not caveat every line; say once which machine the keystrokes are for and point at the keyboard maps.

WARNING

A softmenu page is eighteen items: six unshifted, then six f-shifted, then six g-shifted. Before drawing a

soft key, check the menu array in `src/c47/softmenus.c` and use `mkey`, `fmkey` or `gmkey` to match. Prose like "ADV PGMSLV" reads as unshifted and is wrong: PGMSLV is on the `f` row. Drawing the badges on AN0030 and AN0031 caught two errors that bare command names had hidden.

3.2 Keystroke lines

A keystroke line is set on its own, indented, and wraps at the gaps between keys when it runs long. Two forms. `seq` carries the result the display leaves; `keys` is for a step with nothing to show for it yet.

```
\seq{\key{3}\quad\key{ENTER}\quad\key{4}\quad\key{+}}{X: \ 7}
\keys{\fkey{DISP}\quad\mkey{FIX}\quad\key{4}}
```

3 ENTER 4 + → X: 7

f DISP FIX 4

Separate keys with `\quad` where the reader should pause and with `\` where the keys form one token, as the digits of a number do. `\key{1}\key{1}` is two presses of one number; `\key{1}\quad\key{1}` reads as two separate steps.

Use screen for a display line quoted inside a sentence, like `X: 7`, and keep it to one line. Anything taller belongs in a `prog` block.

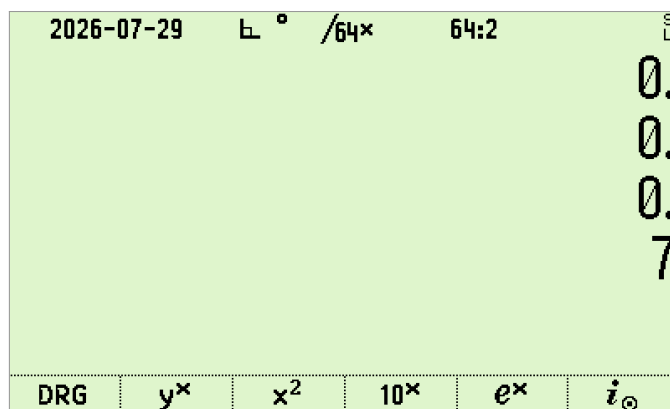
4 A Worked Tutorial

This section is the shape a tutorial should take: state the goal, give the keystrokes, show the screen, say what happened and why. The content is deliberately trivial so the pattern is visible through it.

Add three and four, then set the display to four decimals.

The C47 and R47 use RPN, so there is no equals key. Type the first number, press `ENTER` to separate it from the second, type the second, then press the operation.

3 ENTER 4 + → X: 7

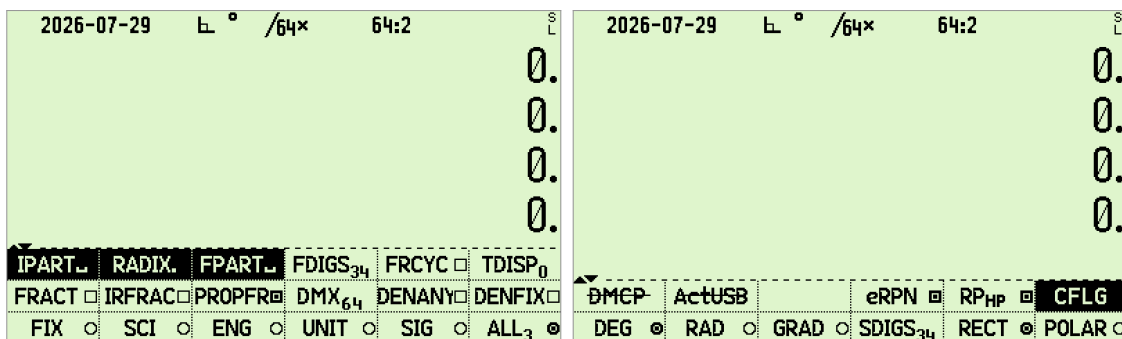


Three plus four. The answer lands in **X**, the bottom line. The tiny **S** over **L** at the top right is the simulator's stack-lift indicator; real hardware carries the battery gauge there instead.

The sum appeared without your saying where to put it. That is the whole of RPN: intermediate results wait on the stack and are consumed by the next operation.

Now the display. Open the display menu with **f** **DISP**. The first row offers the six formats, and **FIX** takes a digit count.

f **DISP** **FIX** **4** → X: 7.0000



Left, the first page of **f** **DISP**: the formats on the unshifted row, the fraction settings on the **f** row above, and **IPART**, **RADIX** and **FPART** on the **g** row. Right, the first page of **f** **MODE** for comparison. A label with a line struck through it, like **DMCP** and **ActUSB** here, is a function the build you are running does not offer.

TIP

Menu paging wraps. A single press of **↓** from page 1 lands on the last page, which is usually quicker than pressing **↑** twice.

Three things make a tutorial section work. Every keystroke line is complete, so a reader can follow it without inferring a missing **ENTER**. Every screen shown is a real capture of exactly those keystrokes. And each step says what happened, not just what to press.

5 Program Entry

A program is given twice: once as a listing, so the reader can check what they have, and once as the keystrokes that produce it. Both, always. A listing alone leaves the reader guessing where MVAR lives; keystrokes alone give them no way to find the step they mistyped.

The listing goes in a prog block, in the display face, with the four-digit step numbers the calculator itself shows:

```
0001: LBL 'HYP'
0002: MVAR 'a'
0003: MVAR 'b'
0004: RCL 'a'
0005: x2
0006: RCL 'b'
0007: x2
0008: +
0009: √x
0010: END
```

Then the keystrokes. Open a new program with `XEQ` `.` `.` and switch to program-entry mode with `f` `PRGM`. Start the label with `g` `LBL`, then `f` `α`, type `H` `Y` `P`, and `ENTER`. MVAR is two pages into the program-functions menu now on screen: `P.FN1`, then `P.FN2`, then `MVAR`, then `f` `α` `a` `ENTER`, and again for `b`. Key the body, then leave program-entry mode with `f` `PRGM`.

NOTE

MVAR is not on `f` `ADV`, which is where readers look for it. In program-entry mode press `P.FN1` on soft key 6, then `P.FN2` on soft key 6 again, then `MVAR` on soft key 4. Say this in full every time; it is the single step readers most often fail to find.

prog also takes anything else that wants the display face over several lines: a matrix as the editor shows it, a menu dump, a two-line result. Rows are separated with `\` and columns lined up with `\`, since the box has no tabbing.

```
y      = 2.1298079
u(y)   = 0.0077526
```

6 Screen Captures

6.1 The rule

Every capture in a note is a real capture from the simulator, driven by the same keystrokes the note prints, started with `--reset` so it reproduces on any machine. Nothing is mocked up and nothing is retouched. The only processing is a threefold point upscale, which keeps the pixels square in print.

This is not a style preference. A note whose pictures were drawn by hand will drift away from the firmware silently, and the reader has no way to tell. A note whose pictures are captures can be regenerated and diffed.

6.2 Placing one

Three macros, by how the picture is used.

Table 5 Capture macros

SOURCE	USE
<code>\shot[w]{file}{caption}</code>	One capture in the flow of the text, centred, not floating, so it stays where the prose put it. Width defaults to 0.58 of the text block.
<code>\shotpair{l}{r}{caption}</code>	Two side by side. Each lands at half the measure, so a pair is for comparing the shape of two screens, not for reading labels off them.
<code>\shotfig[w]{f}{cap}{label}</code>	A numbered float, for a capture the note cross-refers to from elsewhere.

All three prepend `screens/` to the filename and supply the extension, so the argument is bare: `02-result`, not the path and not the `.png`.

Prefer `shot`. A capture belongs beside the keystrokes that produced it, and a float will wander. Reach for `shotfig` only when the note genuinely cross-refers to the picture, as Figure 1 is referred to from Section 13.

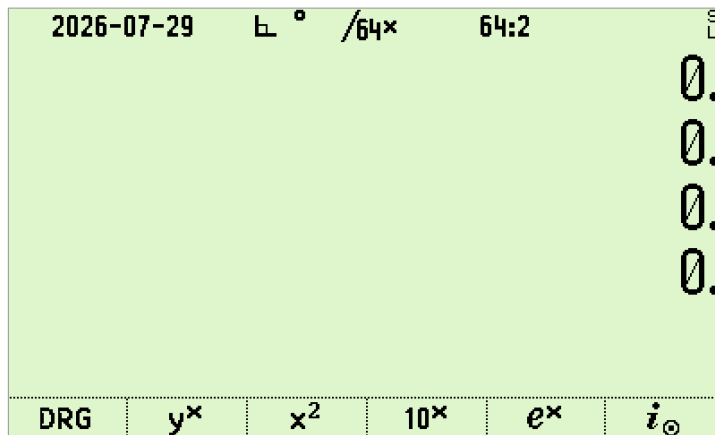


Figure 1 A new calculator, straight out of a reset: an empty stack and MyMenu along the bottom. This is what every capture recipe in `screens/scripts` starts from.

Captions are sentences, in soft grey, and they say what to look at. "The DISP menu" is not a caption. "The formats on the unshifted row, the fraction settings on the f row above" is.

6.3 Reproducing them

Record the recipe. Every note that carries captures carries a `screens/scripts/README.md` giving the build used, the exact command per picture, and the upscale step. The pattern:

```
./c47 --reset --exec 'press "@k 30"; press "@k 12"; press "@k 23";
                    press "@k 36"; snap s2add'
magick s2add.bmp -filter point -resize 300% screens/02-result.png
```

`press "@k NN"` presses physical key NN, numbered row-major from the C47 table in `src/c47/assign.c`. `press @f` and `press @g` set a pending shift, and `press F1` to `press F6` press the soft keys. A longer sequence goes in a `.t47` script beside the README rather than on the command line.

WARNING

A plot needs `--snapskiprefresh`, or `snap` redraws the screen and wipes the drawn pixels. The simulator

also shows a stack-lift indicator at the top right where hardware shows the battery gauge, so say so in the caption the first time a capture appears in a note.

To check a capture set still reproduces after the firmware has moved, rerun the recipe and diff:

```
compare -metric AE /tmp/regenerated.png screens/02-result.png null:
```

Zero differing pixels is the answer you want.

7 Callout Blocks

Three kinds, and three is the whole set. A page carrying six kinds of coloured box carries none, because the reader has stopped distinguishing them.

NOTE

An aside the reader can skip without losing the thread. Background, a cross reference, a reason something is the way it is.

TIP

A shorter route to the same place. A shortcut, a default worth changing, a keystroke that saves three.

WARNING

A trap that will cost the reader an hour. Something that looks right and is not, a step that silently does nothing, or behaviour that differs between the simulator, the DM42, the DM42n and the R47. Spend these sparingly; a note with nine warnings has none.

They are unfilled, bounded on the left only, and titled in a letterspaced label. A rule and a label are enough to set a paragraph apart, and they leave the page quieter than a shaded panel does. A note and a tip carry the same black rule and are told apart by the title alone; only the warning spends the accent, because only the warning is a safety signal.

They break across pages, so a long one is safe. Keep them to a paragraph or two even so: a callout the size of a section should be a section.

The rule for choosing is what the reader loses by skipping it. Nothing, and it is a note. Time, and it is a tip. An hour, and it is a warning.

There is no separate block for hardware differences, and that is deliberate. A feature the reader's build does not carry, a memory ceiling, a capture that will not reproduce: each of those costs an hour, which is the definition of a warning. Name the build in the first sentence and the reader knows whether the box is about them.

8 Code and Text Listings

Two listing styles, both unfilled and closed top and bottom by the same 0.4 pt hairline a capture frame uses. A listing reads as a tabulated block rather than as a shaded panel, which is what keeps it distinct from the prog and screen boxes: those are washed with `lcdbg` because they reproduce the display, and nothing else on the page is.

c47text is the default and takes anything plain: a state-file excerpt, a .t47 script, a shell command, a menu dump.

```
EQUATIONS
1
p×V=n×R×T
OTHER_CONFIGURATION_STUFF
```

mathescape is on, so $\times$ inside a listing renders as the multiplication sign the calculator uses. That is how to quote a formula containing a glyph the keyboard has but ASCII does not.

c47code adds C syntax highlighting and line numbers, for firmware source:

```
1 /* The relative step assumes the model varies on the scale of x. */
2 static real_t centralDifference(real_t (*f)(real_t), real_t x) {
3     const real_t h = fmax(fabs(x), 1.0) * CBRT_EPSILON;
4     return (f(x + h) - f(x - h)) / (2.0 * h);
5 }
```

Quote source sparingly and quote it exactly. A note is not the place to paraphrase the firmware, and a listing that has drifted from the tree is worse than no listing. Give the path with file, as in `src/mathematics/uncertainty.c`, so a reader can go and look.

For one inline fragment use `\lit`, which is `\lit` followed by the text between two matching delimiters. It is the right tool for a macro name, a flag, or a line of a file format.

9 Tables

Tables are booktabs throughout: `\toprule` at the top, one `\midrule` under the head, `\bottomrule` at the bottom. Never a vertical rule and never a double rule, because both of them draw the eye to the grid instead of to the numbers. `@{}` on the outer columns is what makes the rules line up with the text block. Small or footnote size, since a table in body size beside Helvetica prose looks loud. Caption above, labelled TABLE and numbered.

Head cells go in `\hd`, which letterspaces them, so the head reads as furniture without needing a rule or a weight change of its own. To group rows, use `\addlinespace`, never another rule: the palette in Table 3 is grouped that way.

`\table` wraps the whole pattern up, and it is the shortest way to a correct table:

```
\begin{ntable}[@{}l l@{}]{The caption sentence.}{tab:label}
  \hd{Source} & \hd{Means} \\ \midrule
  \lit|\key{7}| & Press that key \\
\end{ntable}
```

Give every table a caption and a label and refer to it in the prose. A table nobody points at is decoration. When a table runs wide, cut `tabcolsep` to 3 pt and drop to `footnotesize` before you consider landscape.

Command reference tables are their own case, and AN0030 is the model: one small table per command, with rows for what it takes from the stack, what it leaves, what it reads, what it writes, and what it refuses. A reader looking up a command wants the same five rows every time, in the same order.

Table 6 The shape of a house table. Three rules, no verticals, head letterspaced.

COLUMN	TYPE	NOTES
First	l	Left, natural width
Second	l	Left, natural width
Third	p{0.42\textwidth}	Fixed width, so prose wraps rather than pushing the table past the margin

10 Charts

Charts are `pgfplots`, wrapped in `nchart` so they float and caption like any other figure. The chrome is recessive by default: axis lines on the left and bottom only, no box, horizontal grid only, ticks outside, labels in `soft`. What is left is the data.

10.1 Why a chart here has almost no colour

This is worth the space, because the conclusion is not a preference. A palette that encodes identity has to clear four things at once: a chroma floor, so a hue does not read as grey; a lightness band; a contrast floor against the paper; and a separation floor, measured pairwise, of $\Delta E \geq 15$ in normal vision and ≥ 8 under simulated protanopia and deuteranopia.

Rule 1 of Section 2.2 has already spent orange on the `f` shift and blue on the `g` shift, and `warnink` has spent red. Testing what is left against those three reserved inks:

Table 7 Every hue that could have been a second series, and what stopped it. Measured pairwise against `warnink`, `fcol` and `gcol`.

CANDIDATE	VERDICT	WHY
violet 8A5CC7	unusable	Collapses against <code>gcol</code> , ΔE 3.4 under protanopia
ochre 8A6B00	unusable	Collapses against <code>warnink</code> , ΔE 2.5 under deuteranopia
teal 00918A	unusable	Chroma 0.089, below the 0.10 floor: reads as grey
teal 00A099	passes	Clears <code>warnink</code> , <code>fcol</code> and <code>gcol</code> on every pair, in normal vision and under all three simulated deficiencies

Exactly one hue survives. It is `chartink`, and a chart gets at most that one, for the single series that is the whole point of the chart. Everything else is `keyink` and the grey ramp.

NOTE

That is not a hardship. A note that will be printed, photocopied and read on a grey electronic-paper screen wants its series told apart by dash pattern and marker anyway, which is also the only encoding that survives a reader with full colour blindness. The constraint and the right answer coincide.

Magnitude uses the grey ramp `seq1` to `seq5`, which passes the ordinal checks: monotone lightness, an adjacent step of at least $\Delta L = 0.06$, and a light end still clearing 2:1 against white. `seq1` is `keyink` and `seq3` is `soft`, so the ramp is built out of the palette rather than beside it.



the magnitude ramp



chartink

10.2 Emphasis: one bar is the point

The commonest chart in this collection ranks contributions and says which one to attack. That is not a categorical chart; it is an *emphasis* chart. One mark in full keyink, the rest in seq4, and the reader's eye goes to the right place without a legend, a hue or a callout.

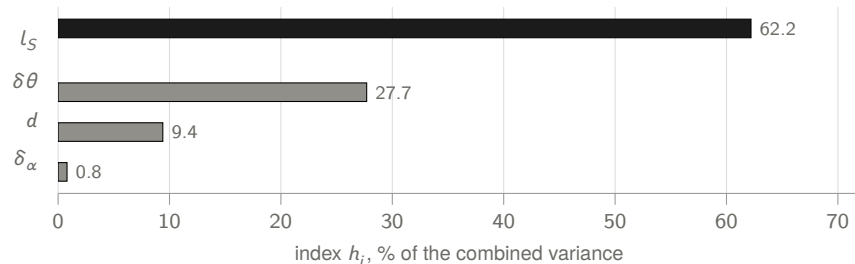


Figure 2 The GUM Annex H.1 uncertainty budget, ranked. l_S carries 62 % of the variance, so it is the only term worth attacking; everything below 10 % is noise. Emphasis, not colour, is what says so.

Sort by magnitude, always, unless the category order carries meaning of its own. An unsorted ranking makes the reader do the sorting.

10.3 One series: the plain case

Most appnote charts have one series, and a single series needs no legend, because the caption names it. Label the axes, give the units, and stop.

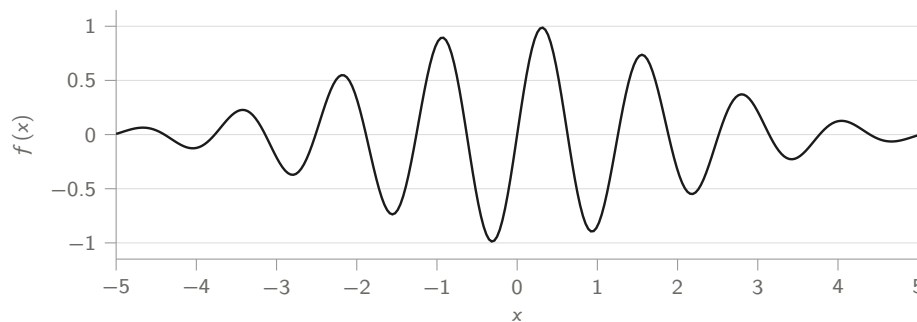


Figure 3 The wave packet $f(x) = \sin(5x) e^{-x^2/8}$ of Application Note 29 [3], over $-5 \leq x \leq 5$. One series, so no legend: the caption names it.

10.4 Two or three series: dash, then marker

Identity runs A, B, C in a fixed order and never cycles: canA solid, canB dashed, canC dotted, all in keyink. A legend appears as soon as there is more than one series. A fourth series is not a fourth dash pattern; it is a second chart, or a small multiple.

Measured data and a fitted curve are a special case of two series that needs no dash ladder, because points and a line are already distinct: use canpoints for the data and canA for the fit.

10.5 The rules, in short

- **Never a second y axis.** Two measures of different scale are two charts, or one chart with both series indexed to a common base. This is the single commonest way to mislead a reader with a chart that is technically accurate.

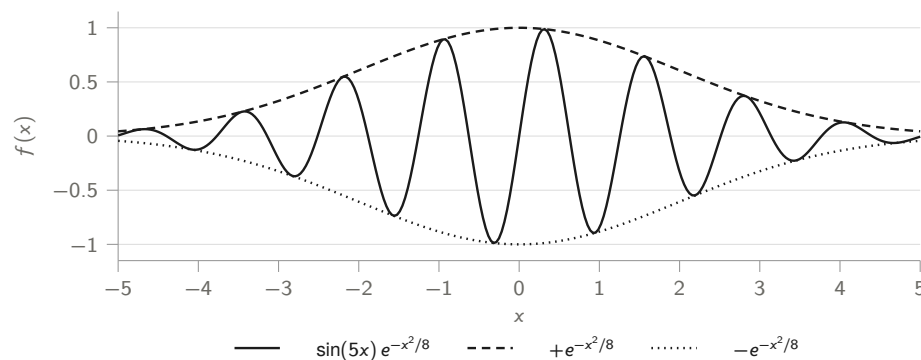


Figure 4 The same packet against the envelope that bounds it, $\pm e^{-x^2/8}$. Solid, dashed and dotted in that fixed order, one ink, and a legend because there is more than one series.

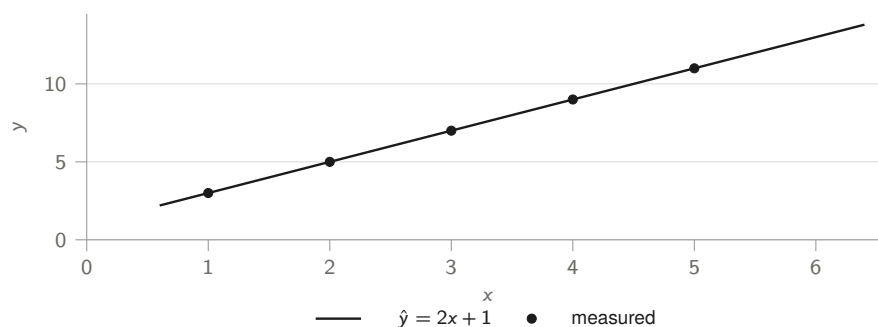


Figure 5 The five pairs of Application Note 29 [3] and the line the regression menu fits to them, $y = 2x + 1$, with $r = 1$. Points against a line need no dash ladder.

- **Identity by dash and marker, in the fixed order A, B, C.** At most one hue, chartink, and only for the series that is the point.
- **Magnitude by the grey ramp,** light to dark, more is darker. Never a rainbow.
- **Emphasis before category.** If the story is "this one dominates", that is one dark mark among grey ones, not five hues.
- **Sort by magnitude** unless the category order means something.
- **Label the axes and give the units.** A number without a unit is not a result.
- **No legend for one series;** a legend for two or more, and direct labels where they fit. Never a value on every point.
- **Recessive chrome.** Horizontal grid only, no box, no fill under a line unless the area is the quantity.

11 Mathematics

Inline mathematics sits in the sentence: the standard uncertainty of an input estimate x_i is written $u(x_i)$, with $\nu_i = n - 1$ degrees of freedom. Display mathematics that is not referred to again goes unnumbered,

$$u(\bar{q}) = \frac{s(q)}{\sqrt{n}}$$

and anything the prose comes back to gets a number and a label:

$$u_c^2(y) = \sum_{i=1}^N c_i^2 u^2(x_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^N c_i c_j u(x_i, x_j) \quad (1)$$

Equation 1 is the law of propagation of uncertainty, and the reference is how the reader gets back to it twelve pages later. Number what you cite and nothing else.

Where a derivation matters, give it. Where only the result matters, give the result and cite the source. A note that reproduces four pages of a standard the reader can download has spent four pages badly.

$$s(q) = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (q_k - \bar{q})^2} \quad (2)$$

$$u_c(y) = \sqrt{\sum_i c_i^2 u^2(x_i)} \quad (3)$$

Units are set upright and separated from the number: 31.71 nm, 50.000838 mm, 0.029 °C. Percent takes a thin space, 95 %, which is the convention the GUM uses and the collection follows.

12 Semantic Markup

Mark up what a thing *is*, not how it should look. Then a later decision about how commands should be set is one edit in the style file instead of a hundred in the notes.

Table 8 Semantic macros

SOURCE	RESULT	FOR
<code>\cmd{U.CALC}</code>	U.CALC	A command as the catalog names it
<code>\vr{U.TAB}</code>	U.TAB	A named variable
<code>\reg{X}</code>	X	A stack register or a labelled display field
<code>\menu{UNCT}</code>	UNCT	A menu
<code>\file{src/c47}</code>	src/c47	A path in the source tree
<code>\swatch{fcol}</code>		A palette chip, outlined so a pale fill still reads
<code>\lsc{note}</code>	NOTE	A letterspaced label: the one typographic accent
<code>\HW{timing}</code>	[HARDWARE: timing]	A figure still to be read off real hardware
<code>\TODO{H.2}</code>	[TODO: H.2]	A hole in the draft

HW and TODO print red on purpose. A draft circulated for comment should make its holes impossible to miss, and a note must not reach master with either still in it. Grep for both before you open the merge request.

13 Cross References and Citations

Label anything you point at and point at it by number, never by position. "Section 7" and "Figure 1" survive an edit; "the section above" and "the figure on page 4" do not.

The label prefixes the collection uses: bare words for sections, as in `\label{callouts}`, and `fig:`, `tab:` and `eq:` for the three numbered kinds.

Cite the other notes. A note that stands alone duplicates one that already exists, and the collection is worth more as a set of cross references than as thirty separate documents. Cite by number in the prose, as in Application Note 29 [3], and give the full title in the bibliography. External standards get a full entry with a URL where one exists [6].

TIP

Before writing a new note, read the ones next to it. AN0029 was nearly filed as redundant with AN0028 until the question was settled: AN0028 tours the settings for a reader who already has opinions about them, and carries no keystrokes or captures at all, while AN0029 starts at the first switch-on. They cite each other now. That conversation belongs in the note's opening paragraphs.

14 Back Matter

In order: any appendices, the keyboard maps if the note gives keystrokes for both machines, the bibliography, then the author block.

`keyboardappendix` emits both maps, each on its own page, labelled `fig:ckbd` and `fig:rkbd` so the body can refer to them. Ninety lines of TikZ used to be pasted into each note that wanted them; now it is one command. Include it in a note that gives keystrokes, and leave it out of one that is purely about the firmware. `ckeyboard` and `rkeyboard` place a single map if you want to lay out the page yourself.

`colophon` takes the firmware version the note was written against and emits a closing page recording what built the document: the three faces and where they come from, the toolchain, the style-file version, and how the captures were made. A note that promises its captures reproduce has to say what they reproduce against, and this is where that promise is recorded.

The author block is `\authorblock{name}{site}{email}` and carries the GFDL notice with it. Every note in the collection is under the GNU Free Documentation License, and both the SPDX line at the top of the source and this block at the bottom are required.

NOTE

The first two lines of every source file are the SPDX identifier and the copyright line, then a comment saying what the note is and how to build it. Copy them from this file; they are what the licence audit reads.

15 Summary

- Copy this directory, rename the `.ltx` per Section 1.2, edit the four front-matter lines, delete the prose.
- Check the author initials against Table 1. There are two Davids.
- Build twice with `TEXINPUTS=../style: lualatex`, under `lualatex`, with `JuliaMono` installed.
- `fcol orange` means the `f` shift and `gcol blue` means the `g` shift. Nothing else may use them.
- Check `src/c47/softmenus.c` before drawing a soft key, and use `mkey`, `fmkey` or `gmkey` to match the row. Eighteen items to a page, six per row.
- Every capture is a real `--reset` capture of the keystrokes printed beside it, with the recipe recorded in `screens/scripts/README.md`.
- Give a program as a listing and as keystrokes. Both.

- Three callout kinds, chosen by what the reader loses in skipping the box. A build difference is a warning, named by build in the first sentence.
- Charts tell series apart by dash and marker, in the fixed order A, B, C. One hue is available, chartink, and exactly one: Table 7 shows what stopped the others. Never a second y axis.
- Label and cite everything you point at, including the other notes.
- Grep for TODO and HW before opening the merge request.

The C47 keyboard

Primary function on the key; orange = f (shift once) above; blue = g (shift twice) below.

$\rightarrow I$ $\Sigma+$ a b/c	y^x $1/x$ #	x^2 $\sqrt{x}$.ms	10^x LOG .d	e^x LN LBL	α XEQ GTO
$ x $ STO $\angle$	% RCL $\Delta\%$	π $R\downarrow$ $\sqrt[y]{x}$	ASIN SIN i	ACOS COS $\rightarrow R$	ATAN TAN $\rightarrow P$
COMPLEX ENTER CPX	LASTx $x\leftrightarrow y$ STK	MODE $+/-$ TRG	DISP EEX EXP	UNDO $\leftarrow$ CLR	
BST $\uparrow$ REGS	EQN 7 HOME	ADV 8 FIN	MATX 9 X.FN	STAT $\div$ PLOT	
SST $\downarrow$ FLGS	BASE 4 BITS	CONV 5 CLK	FLAG 6 REAL	PROB $\times$ INTS	
f/g	ASN 1 KEYS	USER 2 α .FN	P.FN 3 LOOP	PRINT $-$ I/O	
OFF EXIT SNAP	VIEW 0 STOPW	SHOW $\cdot$ INFO	PRGM R/S TEST	CAT $+$ CNST	

The R47 keyboard

The R47 has two physical shift keys: a yellow **f** and a blue **g**. Primary on the key; f above, g below.

<i>i</i> x^2 $\rightarrow R$	<i>i</i>_⊙ $\sqrt{x}$ $\rightarrow P$	<i>x</i>! $1/x$ $.ms$	<i>y</i>^{<i>y</i>} $\sqrt[y]{y}$ $.d$	10^x LOG $\rightarrow I$	<i>e</i>^{<i>x</i>} LN $\#$					
<i> x </i> STO $\nless$	<i>%</i> RCL $\Delta\%$	<i>π</i> $R\downarrow$ $R\uparrow$	USER DRG ASN	<i>f</i>	<i>g</i>					
COMPLEX ENTER CPX						LASTx $x\rightleftharpoons y$ STK	DISP $+/-$ TRG	PFX EEX EXP	UNDO $\leftarrow$ CLR	
<i>α</i> XEQ GTO	SIN 7 ASIN	COS 8 ACOS	TAN 9 ATAN	STAT $\div$ PLOT						
BST $\uparrow$ REGS	BASE 4 BITS	INTS 5 REAL	MATX 6 X.FN	EQN $\times$ ADV						
SST $\downarrow$ FLGS	PREF 1 KEYS	CONV 2 CLK	FLAG 3 <i>α</i>.FN	PROB $-$ FIN						
OFF EXIT INFO	VIEW 0 I/O	SHOW $\cdot$ a b/c	PRGM R/S P.FN	CAT $+$ CNST						

References

- [1] C47/R47 Application Note 018, *Keyboard*.
- [2] C47/R47 Application Note 028, *Configurability Tour*.
- [3] C47/R47 Application Note 029, *First Steps*.
- [4] C47/R47 Application Note 030, *Measurement Uncertainty*.
- [5] C47/R47 Application Note 031, *Algebraic Number Identification*.
- [6] JCGM 100:2008, *Evaluation of measurement data – Guide to the expression of uncertainty in measurement*, Joint Committee for Guides in Metrology, 2008. <https://www.bipm.org/en/committees/jc/jcgm/publications>
- [7] C47/R47 function reference. <https://47calc.com/documentation>
- [8] C47 developer wiki. <https://gitlab.com/rpncalculators/c43/-/wikis/home>
- [9] C47 community wiki. <https://gitlab.com/h2x/c47-wiki>

Colophon

Body and headings	TeX Gyre Heros, the metric-compatible Helvetica from TeX Live
Mathematics	New Computer Modern Sans Math, from TeX Live
Display face	JuliaMono, from juliaweb.org
Typeset with	lua _l atex, TeX Live 2025
Design language	c47appnote v1.1, docs/appnotes/sources/style
Charts	pgfplots, one hue and the grey magnitude ramp
Screen captures	the GTK simulator at --reset, upscaled 3× with a point filter
Written against	firmware 0.109.03.04

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